

Lecture 4. Matrix approach for simple regression model

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Outline

- 1 Matrix review
- 2 Random vectors and matrices
- 3 Simple linear regression model in matrix terms
- 4 Inferences in regression analysis

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Matrix multiplication

- Vector-vector multiplication, dot product, outer product
- Matrix-vector multiplication
- Matrix-matrix multiplication

Matrix review

- Questions 5.1 page 210
- Questions 5.4 page 211
- Questions 5.14 page 211
- Questions 5.19 page 211

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Expectation of Random Vector or Matrix

- A random vector or a random matrix contains elements that are random variables.
- Observations vector $\mathbf{Y}$, error terms vector ϵ

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

- Expected value of $\mathbf{Y}$ is a vector:

$$E\{\mathbf{Y}\} = \begin{bmatrix} E\{Y_1\} \\ E\{Y_2\} \\ \vdots \\ E\{Y_n\} \end{bmatrix} \quad E\{\epsilon\} = \begin{bmatrix} E\{\epsilon_1\} \\ E\{\epsilon_2\} \\ \vdots \\ E\{\epsilon_n\} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Variance-covariance matrix of random vector

The variance-covariance matrix of $\mathbf{Y}$:

$$\begin{aligned}\text{var}\{\mathbf{Y}\} &= \begin{bmatrix} \text{var}\{Y_1\} & \text{cov}\{Y_1, Y_2\} & \dots & \text{cov}\{Y_1, Y_n\} \\ \text{cov}\{Y_2, Y_1\} & \text{var}\{Y_2\} & \dots & \text{cov}\{Y_2, Y_n\} \\ \dots & \dots & \dots & \dots \\ \text{cov}\{Y_n, Y_1\} & \text{cov}\{Y_n, Y_2\} & \dots & \text{var}\{Y_n\} \end{bmatrix} \\ &= E\{[\mathbf{Y} - E\{\mathbf{Y}\}][\mathbf{Y} - E\{\mathbf{Y}\}]'\},\end{aligned}$$

where

- $\text{var}\{Y_i\}$ is variance of Y_i

$$\text{var}\{Y_i\} = E\{(Y_i - E\{Y_i\})^2\},$$

- $\text{cov}\{Y_i, Y_j\}$ is covariance of Y_i, Y_j :

$$\text{cov}\{Y_i, Y_j\} = E\{(Y_i - E\{Y_i\})(Y_j - E\{Y_j\})\}.$$

Variance-covariance matrix of random vector

- $\text{var}\{\mathbf{Y}\}$ is a symmetric matrix.
- In normal error regression model, variance-covariance matrix for the random vector ϵ is

$$\text{var}\{\epsilon\} = \sigma^2 I,$$

where σ^2 is the common variance of error term.

- Denote random matrix $\mathbf{W} = \mathbf{A}\mathbf{Y}$ where $\mathbf{Y}$ is random vector and $\mathbf{A}$ is a constant matrix.

$$\begin{aligned} E\{\mathbf{W}\} &= \mathbf{A}E\{\mathbf{Y}\} \\ \text{var}\{\mathbf{W}\} &= \mathbf{A}\text{var}\{\mathbf{Y}\}\mathbf{A}' \end{aligned}$$

Questions

- See example page 196
- Questions 5.17 page 211

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Simple linear regression model in matrix terms

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

For normal error regression model, $\boldsymbol{\epsilon}$ is a vector of independent normal random variables with $E\{\boldsymbol{\epsilon}\} = \mathbf{0}$ and $\text{var}\{\boldsymbol{\epsilon}\} = \sigma^2 \mathbf{I}$.

Expected values

$$E\{\mathbf{Y}\} = \mathbf{X}\boldsymbol{\beta}.$$

Least squares estimation of regression parameters

- The solution of least square satisfies normal equation (pls. read textbook for the derivation):

$$\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{Y}$$

where $\mathbf{b}$ is the vector of the least squares regression coefficients:

$$\mathbf{b} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

- Assume that matrix $\mathbf{X}'\mathbf{X}$ is invertible. Multiply both sides with $(\mathbf{X}'\mathbf{X})^{-1}$, we obtain

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Toluca example

- Show Toluca example, page 200 (in-class work)

Fitted values (mean responses)

Fitted vector:

$$\begin{aligned}\hat{\mathbf{Y}} &= \begin{bmatrix} \hat{Y}_1 \\ \hat{Y}_2 \\ \vdots \\ \hat{Y}_n \end{bmatrix} = \mathbf{X}\mathbf{b} \\ &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= \mathbf{H}\mathbf{Y},\end{aligned}$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ called *hat matrix*. It plays an important role in diagnostics for regression analysis as we shall see later.

Hat matrix is symmetric and idempotency: $\mathbf{H}\mathbf{H} = \mathbf{H}$.

Residuals

- Residuals $e_i = Y_i - \hat{Y}_i$.

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\mathbf{b} \\ = (\mathbf{I} - \mathbf{H})\mathbf{Y}$$

- Variance-covariance matrix of residuals:

$$\begin{aligned} \text{var}\{\mathbf{e}\} &= (\mathbf{I} - \mathbf{H})\text{var}\{\mathbf{Y}\}(\mathbf{I} - \mathbf{H})' \\ &= \sigma^2(\mathbf{I} - \mathbf{H})\mathbf{I}(\mathbf{I} - \mathbf{H}) \\ &= \sigma^2(\mathbf{I} - \mathbf{H}) \end{aligned}$$

and is estimated by:

$$s^2\{\mathbf{e}\} = \text{MSE} * (\mathbf{I} - \mathbf{H})$$

Example

- Show Toluca example, pages 202, 203 (in-class work)

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ANOVA sum of square

- The total sum of squares:

$$\begin{aligned}SSTO &= \sum (Y_i - \bar{Y})^2 = \sum Y_i^2 - \frac{(\sum \bar{Y})^2}{n} \\&= \mathbf{Y}'\mathbf{Y} - \frac{\mathbf{Y}'\mathbf{J}\mathbf{Y}}{n},\end{aligned}$$

where $\mathbf{J}$ is the matrix of 1s: $\mathbf{J} = \begin{bmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{bmatrix}$

- Sum of square error

$$SSE = \mathbf{Y}'\mathbf{Y} - \mathbf{b}'\mathbf{X}'\mathbf{Y}$$

- Sum of square residuals:

$$SSR = SSTO - SSE = \mathbf{b}'\mathbf{X}'\mathbf{Y} - \frac{\mathbf{Y}'\mathbf{J}\mathbf{Y}}{n}$$

- Double check with Toluca company example, page 205 (in-class work).

ANOVA sums of squares as quadratic forms

Quadratic form: $\mathbf{Y}'\mathbf{A}\mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} Y_i Y_j$, which is a scalar value. We rewrite sum of squares in quadratic forms as follows:

$$SSTO = \mathbf{Y}' \left[\mathbf{I} - \left(\frac{1}{n} \right) \mathbf{J} \right] \mathbf{Y}$$

$$SSE = \mathbf{Y}'(\mathbf{I} - \mathbf{H})\mathbf{Y}$$

$$SSR = \mathbf{Y}' \left[\mathbf{H} - \left(\frac{1}{n} \right) \mathbf{J} \right] \mathbf{Y},$$

where $\mathbf{I} - \left(\frac{1}{n} \right) \mathbf{J}$, $\mathbf{I} - \mathbf{H}$, and $\mathbf{H} - \left(\frac{1}{n} \right) \mathbf{J}$ are $n \times n$ symmetric matrices.

- Observations from previous lectures:
 - All interval estimates are of the following form: point estimator plus and minus a certain number of estimated standard deviations of the point estimator.
 - All tests require the point estimator and the estimated standard deviation of the point estimator or, in the case of analysis of variance tests, various sums of squares
- Hence, we focus here chiefly on the matrix expressions for the estimated variances of point estimators of interest

- The variance-covariance matrix of $\mathbf{b}$:

$$\begin{aligned}\text{var}\{\boldsymbol{\beta}\} &= \begin{bmatrix} \text{var}\{\beta_0\} & \text{cov}\{\beta_0, \beta_1\} \\ \text{cov}\{\beta_0, \beta_1\} & \text{var}\{\beta_1\} \end{bmatrix} \\ &= \sigma^2 * (\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

where σ^2 is the common variance of error terms.

- When MSE is substituted for σ^2 , we obtain the estimated variance-covariance matrix of $\mathbf{b}$, denoted by $s^2\{\mathbf{b}\}$:

$$s^2\{\mathbf{b}\} = \text{MSE} * (\mathbf{X}'\mathbf{X})^{-1}$$

Toluca example

- Show Toluca example, page 207 (in-class work)

Mean response

- Mean response at $\mathbf{X}_h = \begin{bmatrix} 1 \\ X_h \end{bmatrix}$:

$$\hat{Y}_h = \mathbf{X}_h' \mathbf{b}.$$

- The variance of $\hat{Y}_h$ is:

$$\text{var}\{\hat{Y}_h\} = \sigma^2 * \mathbf{X}_h' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}_h$$

- The estimated variance of $\hat{Y}_h$ is:

$$s^2\{\hat{Y}_h\} = \text{MSE} * (\mathbf{X}_h' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}_h)$$

Toluca example

- Show Toluca example, page 208 (in-class work)

- Exercise 5.23, page 212 (in-class work)

Summary

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- 1 Chapter 5, Applied Linear Regression Models, Michael H. Kutner *et al.*, 2004